Department of Computer Technology and Information Systems

CTIS 365: Applied Data Analysis

Semester: Spring 2025-2026

Lab Guide 5

OBJECTIVES: Z-Score, Data Manipulation with dplyr
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Q1. Create the vector with following data (67,34,54,23,45,67,89,60,34,56,99,76,86,32,15).

a. Transform all scores to z-scores.

b. Find the IQ score for every participant.

c. Store the IQ scores into the vector.

d. Display the first 6 rows of the final data set.

Hint: To calculate the IQ score for one participant use the below formula

IQ.score <- 100 + (data * 15)

Q2. Read **countries.csv** file. (Do not forget to include dplyr library and use %>% operator)

a) Read the file to a data set named counties. Select the following columns to a new data set countries_selected;

- State
- Metro
- Region
- Income
- Unemployment
- Citizens
- Population
- Walk

```
head(counties_selected)
   metro region   state income unemployment citizens population walk
   Metro  South Alabama  51281          7.6    40725      55221  0.5
   Metro  South Alabama  50254          7.5   147695     195121  1.0
Nonmetro  South Alabama  32964         17.6    20714      26932  1.8
   Metro  South Alabama  38678          8.3    17495      22604  0.6
   Metro  South Alabama  45813          7.7    42345      57710  0.9
Nonmetro  South Alabama  31938         18.0     8057      10678  5.0
```

b) Count the number of countries in each state weighted based on the citizens.

```
            state        n
1         Alabama  3620994
2          Alaska   519266
3         Arizona  4526594
4        Arkansas  2164083
5      California 24280349
6        Colorado  3750953
7     Connecticut  2574178
8        Delaware   681606
9         Florida 13933052
10        Georgia  6978660
11         Hawaii  1001729
```

c) Find the US states where the most people walk to work.

- Use walk column.
- Calculate and add a column called pop_walk containing the total number of people who walk to work in the each county. It is calculated by walk_population=population*walk/100.
- Use a (weighted and sorted) count() function to find the total number of people who walk to work in each state.

d) Group the data by state and summarize to create the columns named as maxUnemployment and avgIncome.

```
A tibble: 50 x 3
   state       maxUnempl averageIncome
   <chr>           <dbl>         <dbl>
 1 Alabama          22.6        37973.
 2 Alaska           21.9        65125.
 3 Arizona          19.8        43252.
 4 Arkansas         18.1        36626.
 5 California       17.4        56013.
 6 Colorado         27          51263.
 7 Connecticut       9.5        71184.
 8 Delaware          8.4        58068.
 9 Florida          15.8        44046.
10 Georgia          20.6        40705.
... with 40 more rows
```

e) Calculate the average income of countries within each region and state. Find the average income state in each region.

```
# Groups:   region [4]
   region        state        avgIncome
   <chr>         <chr>            <dbl>
 1 North Central Illinois        50163.
 2 North Central Indiana         48745.
 3 North Central Iowa            50483.
 4 North Central Kansas          47322.
 5 North Central Michigan        44465.
 6 North Central Minnesota       53927.
 7 North Central Missouri        41755.
 8 North Central Nebraska        48646.
 9 North Central North Dakota    55575.
10 North Central Ohio            48446.
```

f) Find the total population for each combination of state and metro.

```
Groups:   state [50]
   state      metro    total_Pop
   <chr>      <chr>        <int>
 1 Alabama    Metro      3671377
 2 Alabama    Nonmetro   1159243
 3 Alaska     Metro       494990
 4 Alaska     Nonmetro    230471
 5 Arizona    Metro      6295145
 6 Arizona    Nonmetro    346783
 7 Arkansas   Metro      1806867
 8 Arkansas   Nonmetro   1151341
 9 California Metro     37587429
10 California Nonmetro    834035
```